

Airframe Parameter Comparison

All FC parameters in unified-float format (radians, volts, fractions). **Bold** values are dramatic outliers relative to the airframe set median.

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
RollRateKp	0.1	0.15	0.24	0.2	0.3	0.095	0.16	0.095	0.16
AltPosKi	0.0046	0.003	0.004	0.003	0.004	0.00276	0.00092	0.00184	0.0138
RollAngleKp	7	7	8	7	8	5.75	7	5.75	7
RollAngleIntLimit	0.002618	0.003	0.005	0.003	0.005	0.002618	0.00733	0.002618	0.005236
PitchRateKp	0.1	0.18	0.28	0.22	0.32	0.095	0.16	0.095	0.16
AltPosKp	0.5124	0.45	0.5	0.55	0.65	0.4392	0.1464	0.4575	0.915
PitchAngleKp	7	7	8	7	8	5.75	7	5.75	7
PitchAngleIntLimit	0.002618	0.003	0.005	0.003	0.005	0.002618	0.00733	0.002618	0.005236
YawRateKp	0.1	0.18	0.26	0.15	0.25	0.125	0.2	0.125	0.2
RollRateKd	0.0045	0.012	0.008	0.006	0.004	0.004	0.006	0.004	0.006
BbLogType	0	0	0	0	0	0	0	0	0
RxThrottleCh	0	0	0	0	0	2	0	0	0
LowVoltThres	9.6	9.6	9.6	9.6	9.6	10	13.6	13.8	3
RollCamKp	1	1	1	1	1	1	1	1	1
EstCruiseThr	0.54	0.55	0.55	0.6	0.6	0.5	0.31	0.3	0.54
StickHysteresis	0.02	0.02	0.02	0.02	0.02	0.02	0.04	0.03	0.02
FwClimbThrottle	0	0	0	0	0	0	0	0	0
PercentIdleThr	0.05	0.05	0.05	0.05	0.05	0.1	0.1	0.05	0.1
RollAngleKi	0.25	0.25	0.2	0.25	0.2	0.15	0.2	0.15	0.1
PitchAngleKi	0.25	0.25	0.2	0.25	0.2	0.15	0.1	0.15	0.1
PitchCamKp	1	1	1	1	1	1	1	1	1
ServoLpfHz	25	25	25	25	25	25	10	25	25
PitchRateKd	0.0045	0.012	0.008	0.008	0.005	0.004	0.006	0.004	0.006
NavVelKp	0.3	0.25	0.35	0.3	0.4	0.3	0.6	0.6	0.6
Horizon	0.3	0.3	0.5	0.3	0.5	0.4	0.4	0.4	0.3
MadgwickKpMag	0.5	0.5	0.5	0.5	0.5	0	0.5	0.4	0.5
NavRthAlt	10	10	10	10	10	5	10	10	10
NavMagVar	0.205949	0.223402	0.223402	0.223402	0.223402	0.059341	0.226893	0.204204	0.226893
RcChannels	7	7	7	7	7	7	7	7	7
RxRollCh	1	0	0	0	0	0	1	1	1
MadgwickKpAcc	0.2	0.2	0.25	0.2	0.25	0.05	0.2	0.18	0.2
RollCamTrim	0	0	0	0	0	0	0.01	0	0
NavPosIntLimit	3	5	6	5	6	3	5	5	8
RxPitchCh	2	0	0	0	0	1	2	2	2
RxYawCh	3	0	0	0	0	3	3	3	3
MaxDescentRateDmpS	10	10	10	10	10	9	18	30	12
DescentDelayS	15	15	15	15	15	10	10	10	15
GyroLpfSel	2	2	1	2	1	2	2	2	2
NavCrossTrackKp	0.04	0.04	0.05	0.04	0.05	0.04	0.04	0.04	0.04
RxGearCh	4	0	0	0	0	5	5	5	5
RxAux1Ch	5	0	0	0	0	6	8	11	8
ServoSense	0	0	0	0	0	0	0	0	9

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
AccConfSd	0.06	0.06	0.06	0.06	0.06	0.02	0.06	0.04	0.06
BatteryCapacity	22	2200	2200	2200	2200	50	22	22	22
RxAux2Ch	6	0	0	0	0	4	6	6	6
RxAux3Ch	10	0	0	0	0	8	7	7	7
NavPosKp	0.2475	0.22	0.28	0.25	0.3	0.2475	0.2475	0.2475	0.33
AltLpf	10	10	10	10	10	10	20	10	20
Balance	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5
RxAux4Ch	8	0	0	0	0	11	11	8	10
NavPosKi	0.012	0.012	0.015	0.012	0.015	0.012	0.012	0.012	0.008
TiltThrottleFf	0	0	0	0	0	0.07	0.25	0.1	0
MaxYawRate	2.0944	1.5708	3.14159	1.5708	3.14159	3.49066	7.67945	5.23599	7.67945
FwRollPitchFf	0	0	0	0	0	0	0	0	0.25
FwPitchThrottleFf	0	0	0	0	0	0	0	0	0.1
FwMaxClimbAngle	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	1.0472	0.261799
NavMaxAngle	0.436332	0.436332	0.523599	0.436332	0.523599	0.261799	0.349066	0.349066	0.261799
FwSpoilerDecayPercent	0.01	0.01	0.01	0.01	0.01	0	0	0	0.015
FwAileronDifferential	0	0	0	0	0	0	0	0	0.3
KfAccUBiasVar	3e-06	0	0	0	0	1e-05	5e-06	0	0
MaxPitchAngle	0.523599	0.523599	0.785398	0.523599	0.785398	1.0472	1.0472	1.0472	1.0472
QuatGain	-	4	4	4	4	-	-	-	-
MaxRollAngle	0.523599	0.523599	0.785398	0.523599	0.785398	1.0472	1.0472	1.0472	1.0472
YawLpfHz	50	50	60	50	60	50	50	50	30
NavHeadingTurnout	0.349066	0.349066	0.349066	0.349066	0.349066	0.349066	0.349066	0.349066	0.698132
AltHoldThrCompDecayPercent	0.03	0.03	0.03	0.03	0.03	0	0	0.001	0
FwBoardPitchAngle	0	0	0	0	0	0	0	0	0.087266
MaxRollRate	10.472	6.28319	10.472	6.28319	10.472	10.472	10.472	10.472	10.472
MaxPitchRate	10.472	3.14159	6.28319	3.14159	6.28319	10.472	10.472	10.472	10.472
CurrentScale	100	100	100	100	100	100	100	100	100
VoltScale	100	100	100	100	100	100	100	100	100
FwAileronRudderMix	0	0	0	0	0	0	0	0	1
FwAltSpoilerFf	0	0	0	0	0	0	0	0	0.5
MaxCompassYawRate	1.0472	0.523599	0.523599	0.523599	0.523599	1.0472	1.0472	1.0472	1.5708
AccLpfSel	4	4	3	4	3	4	4	4	4
YawRateKd	0.001125	0.008	0.005	0.002	0.002	0.001125	0.001125	0.001125	0.000125
ThrottleGainRate	0	0	0	0	0	0	0	0	0
RxAux5Ch	9	0	0	0	0	9	9	9	9
RxAux6Ch	7	0	0	0	0	10	10	10	11
RxAux7Ch	11	0	0	0	0	7	4	4	4
YawAngleKp	3	6	8	4	5	3	3	3	2.25
YawAngleKi	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.25
YawAngleIntLimit	0.008727	0.025	0.03	0.015	0.02	0.008727	0.008727	0.008727	0.008727
AltPosIntLimit	0.5	0.25	0.3	0.35	0.4	0.5	0.5	0.5	0.5
MotorStopSel	0	0	0	0	0	0.002618	0	0	0
AltThrottleCompLimit	0.2	0.25	0.28	0.22	0.24	0.06	0.12	0.05	0.02
VrsRoc	50	-3	-3	-3	-3	30	30	30	50
AhRocWindowMps	20	1	1	1	1	15	30	5	5

Param	Ecks 220	Ecks 0800 Mod	Ecks 0800 Sport	Ecks 1kg Mod	Ecks 1kg Sport	Rok's Quad	Ken's 450	Ken's Alpha	Ken's LadyBug
NavProxAltM	4	4	4	4	4	2	3	3	5
NavProxRadiusM	3	3	3	3	3	2	2	1	10
KfBaroVar	100	1	1	1	1	20	50	50	50
KfAccUVar	0.2	2	2	2	2	0.2	0.2	0.2	0.2
FwStickScale	4	0.4	0.5	0.4	0.5	4	2	4	4
FwRollControlPitch	0.785398	45.8366	45.8366	45.8366	45.8366	0.785398	0.785398	0.785398	0.785398
AhThrottleMovingTrigger	0.2	0.2	0.15	0.2	0.15	0.1	0.11	0.2	0.3
NavFenceRadiusM	100	400	400	400	400	100	200	100	100
DiveRecoverAlt	-	0	0	0	0	-	-	-	-

Notes

- Outlier detection: values > 3 MAD from the median are bolded.
- Unused/reserved parameters omitted.